THE UNIVERSITY OF SYDNEY

MOOC INTRODUCTION TO CALCULUS

Notes for "Optimisation"

Examples:

1. Suppose we want to minimise the result of summing a positive real number with its reciprocal. One can experiment with a few numbers to get a feel for the problem, for example,

$$2+\frac{1}{2}=\frac{5}{2}\,, \qquad 10+\frac{1}{10}=\frac{101}{10}\,, \qquad 1+\frac{1}{1}=2\,,$$
$$0.5+\frac{1}{0.5}=\frac{1}{2}+2=\frac{5}{2}\,, \qquad 0.1+\frac{1}{0.1}=\frac{1}{10}+10=\frac{101}{10}\ .$$

Just by experimentation, one can quickly become convinced that 2 is the smallest such sum (when one adds 1 to its reciprocal, which is also 1), but how can one be sure?

Consider the function

$$y \;=\; f(x) \;=\; x+\frac{1}{x} \;=\; x+x^{-1}\ .$$

The strategy is to use the derivative to find the turning point. Observe that

$$y' \;=\; 1-x^{-2} \;=\; 1-\frac{1}{x^2} \;=\; \frac{x^2-1}{x^2}\ ,$$

so that $y'=0$ precisely when the numerator $x^2-1=0$, that is, when $x=\pm 1$. Note that y' is undefined for $x=0$. We get the following sign diagram:

x		-1		0		1	
y'	$+$	0	$-$	u	$-$	0	$+$

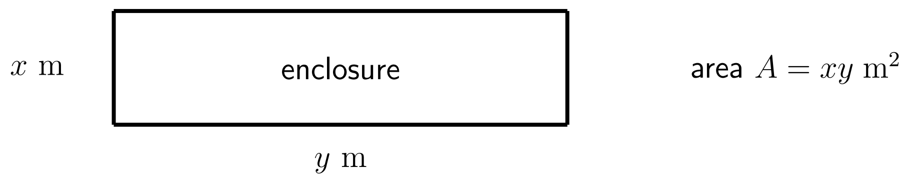
This indicates that the curve is increasing for $x<-1$, decreasing for $-1<x<0$, decreasing again for $0<x<1$ and increasing for $x>1$, with a maximum at $x=-1$ and a minimum at $x=1$.

However, the application to our original problem only uses $x>0$, and so we get the minimum sum of a positive number with its reciprocal to be

$$f(1) \;=\; 1+\frac{1}{1} \;=\; 2\ .$$

2. Suppose one wishes to find the maximum area one can make with a rectangular enclosure using exactly 100 metres of fencing. If one experiments with a few diagrams and figures, one will quickly start to convince oneself that forming a square with equal side-lengths 25 metres appears to achieve the maximum area, which should therefore be $25^2=625$ square metres. How can one be sure one is unable do any better than that?

Here is a diagram of an enclosure in the shape of a rectangle with side lengths x metres and y metres and area $A = xy$ square metres:



We are in fact expecting the final answer to use a square enclosure, so this diagram is only a guide to the mathematics that follows. The constraint is that we are using 100 metres of fencing, so that, adding up all four side lengths, we get

$$100 \;=\; x + x + y + y \;=\; 2x + 2y \;=\; 2(x+y) \;,$$

so that

$$x + y \;=\; 50 \qquad \text{and} \qquad y \;=\; 50 - x \;.$$

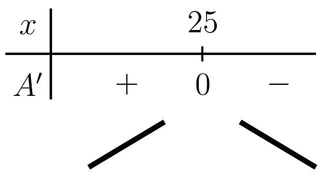
We can now regard A as a function of x only, giving

$$A \;=\; xy \;=\; x(50 - x) \;=\; 50x - x^2 \;.$$

Its derivative with respect to x is

$$A' \;=\; 50 - 2x \;,$$

which is zero precisely when $x = 25$. We get the following sign diagram for A':



This indicates that the area function A is increasing for $x < 25$ and decreasing for $x > 25$, attaining a global maximum

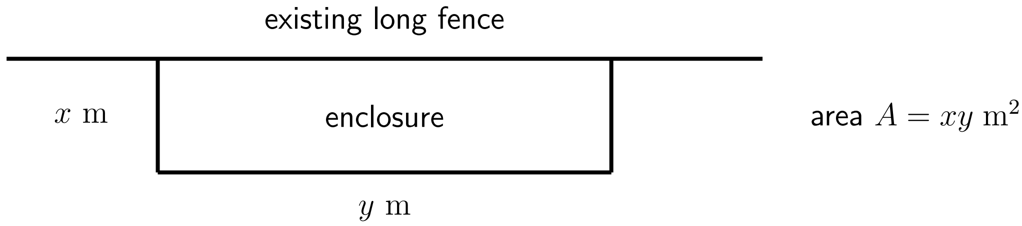
$$A \;=\; 25(50 - 25) \;=\; 625$$

when $x = 25$. Thus the maximum area that can be achieved by the rectangular enclosure is 625 square metres, and this occurs when $x = y = 25$, so that the enclosure is indeed square.

(Note that x is nonnegative by the physical constraint of the problem, though this maximum also holds when A is regarded as a quadratic function for all real x. The global maximum corresponds to the apex of an inverted parabola.)

3. Suppose we modify the previous problem and again form a rectangular enclosure using 100 metres of fencing, but now we have an existing long fence that we can use as one of the side lengths. The problem again is maximise the area of the enclosure.

Denote the length of the two new sides perpendicular to the existing fence by x m, and the length of the new side parallel to the existing fence by y m. The area again is denoted by $A = xy$ square metres.



The fencing constraint now tells us that $100 = 2x + y$, so that $y = 100 - 2x$. Hence

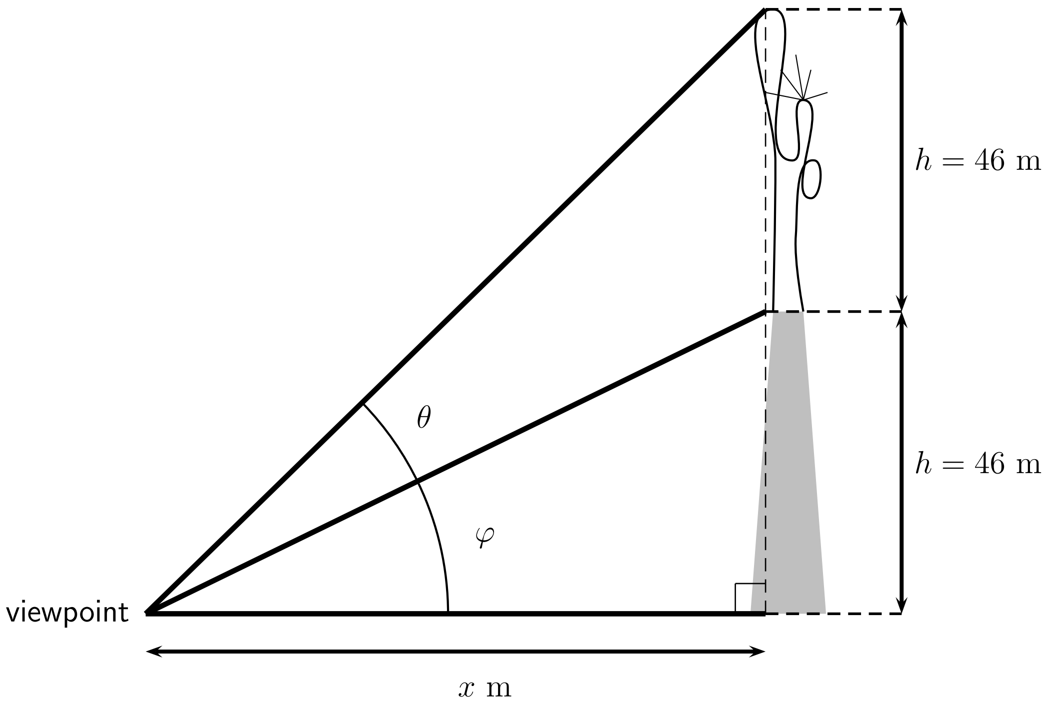
$$A \ = \ xy \ = \ x(100 - 2x) \ = \ 100x - 2x^2 \ .$$

Now we have $A' = 100 - 4x$, which is zero precisely when $x = 25$ (as in our previous example), which occurs when $y = 50$ (so the corresponding rectangle is not a square, which differs from our previous example). The sign diagram for A' is the same as in the previous solution, confirming that a global maximum occurs when $x = 25$, namely

$$A \ = \ 25 \times 50 \ = \ 1,250 \ .$$

Thus the largest area that can be achieved is $1,250$ square metres, by using the 100 metres of fencing to form two new sides perpendicular to the existing fence, each of length 25 metres, and one new side parallel to the existing fence, of length 50 metres.

4. In this final example, we want to find the maximum angle subtended by the Statue of Liberty, as well as the horizontal distance from the base of the Statue that achieves this maximum angle. Note that the Statue and its pedestal are both $h = 46$ metres tall.



We therefore want to maximise the angle θ shown in the above diagram, when viewing the Statue from a horizontal distance of x metres. Let φ be the angle subtended by the pedestal. By inspecting right-angled triangles, we have

$$\tan\varphi \;=\; \frac{h}{x} \qquad \text{and} \qquad \tan(\theta+\varphi) \;=\; \frac{2h}{x}\;.$$

Hence

$$\varphi \;=\; \tan^{-1}\left(\frac{h}{x}\right) \qquad \text{and} \qquad \theta+\varphi \;=\; \tan^{-1}\left(\frac{2h}{x}\right)\;,$$

so that

$$\theta \;=\; \tan^{-1}\left(\frac{2h}{x}\right) - \varphi \;=\; \tan^{-1}\left(\frac{2h}{x}\right) - \tan^{-1}\left(\frac{h}{x}\right)\;.$$

Looking at the derivative of θ with respect to x is likely to lead to progress in determining the maximum possible value of θ. But the derivative of a difference is the difference of the derivatives, so that, by the Chain Rule,

$$\frac{d\theta}{dx} \;=\; \frac{d}{du}\big(\tan^{-1} u\big)\frac{du}{dx} - \frac{d}{dv}\big(\tan^{-1} v\big)\frac{dv}{dx}$$

where $u = \frac{2h}{x} = 2hx^{-1}$ and $v = \frac{h}{x} = hx^{-1}$, so that

$$\frac{du}{dx} \;=\; -2hx^{-2} \;=\; \frac{-2h}{x^2} \qquad \text{and} \qquad \frac{dv}{dx} \;=\; -hx^{-2} \;=\; \frac{-h}{x^2}\;.$$

But we know that

$$\frac{d}{dx}(\tan^{-1} x) \;=\; \frac{1}{x^2+1}$$

(from an earlier video where we introduced the *witch of Maria Agnesi*). Putting all of these ingredients together, we get

$$\begin{aligned}
\frac{d\theta}{dx} \;&=\; \left(\frac{1}{u^2+1}\right)\left(\frac{-2h}{x^2}\right) - \left(\frac{1}{v^2+1}\right)\left(\frac{-h}{x^2}\right) \\[2ex]
&=\; \left(\frac{1}{\frac{4h^2}{x^2}+1}\right)\left(\frac{-2h}{x^2}\right) + \left(\frac{1}{\frac{h^2}{x^2}+1}\right)\left(\frac{h}{x^2}\right) \\[2ex]
&=\; \left(\frac{x^2}{4h^2+x^2}\right)\left(\frac{-2h}{x^2}\right) + \left(\frac{x^2}{h^2+x^2}\right)\left(\frac{h}{x^2}\right) \\[2ex]
&=\; \frac{-2h}{4h^2+x^2} + \frac{h}{h^2+x^2} \;=\; \frac{-2h(h^2+x^2)+h(4h^2+x^2)}{(4h^2+x^2)(h^2+x^2)} \\[2ex]
&=\; \frac{h(-2h^2-2x^2+4h^2+x^2)}{(4h^2+x^2)(h^2+x^2)} \;=\; \frac{h(2h^2-x^2)}{(4h^2+x^2)(h^2+x^2)}\;.
\end{aligned}$$

Thus we get

$$\frac{d\theta}{dx} \;=\; \frac{h(2h^2-x^2)}{(4h^2+x^2)(h^2+x^2)}\;,$$

and notice, on the right-hand side, that h and the denominator are always positive.

In what follows we make the underlying assumption throughout that x is positive (which accords with the physical constraints of the original problem involving the Statue). Hence $\frac{d\theta}{dx} = 0$ precisely when $2h^2 = x^2$, that is, $x^2 = 2h^2$, so that

$$x \;=\; \sqrt{2}\,h\;.$$

We can build the following sign diagram for $\frac{d\theta}{dx}$ (only considering $x > 0$), noting that the sign is completely determined by the expression $2h^2 - x^2$:

x		$\sqrt{2}\,h$	
$\frac{d\theta}{dx}$	$+$	0	$-$

Hence θ is increasing for $x < \sqrt{2}\,h$ and decreasing for $x > \sqrt{2}\,h$, achieving a global maximum when the horizontal distance to the Statue is

$$x \;=\; \sqrt{2}\,h \;=\; 46\sqrt{2} \;\approx\; 65\,\mathrm{m}\;.$$

The maximum angle then becomes

$$\theta \;=\; \tan^{-1}\left(\frac{2h}{\sqrt{2}\,h}\right) - \tan^{-1}\left(\frac{h}{\sqrt{2}\,h}\right) \;=\; \tan^{-1}(\sqrt{2}) - \tan^{-1}\left(\frac{1}{\sqrt{2}}\right) \;\approx\; 19.5^\circ\;.$$

Notice that the final answer for θ is independent of the height h of the Statue and its pedestal. We reach the same answer for the maximum angle subtended by any statue that is mounted on a pedestal of the same height.

The distance $\sqrt{2}\,h$ that achieves this maximum angle of course varies with h. In our problem we achieve a maximum angle that is slightly less than 20 degrees and at a horizontal distance from the Statue of about 65 metres.